

EV CHARGING STATION CONTROLLER USING VERILOG HDL

VLSI Skill Lab Project Report

Submitted by:

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1. Introduction

Electric Vehicles (EVs) have become an important part of modern transportation due to their environmental benefits and lower operating costs. An EV charging station requires a controller that ensures charging begins only under safe and valid conditions.

This project implements an EV Charging Station Controller using Verilog HDL. The controller uses a Moore Finite State Machine (FSM) to manage the charging process by monitoring vehicle connection, authentication, battery status, and disconnection events. The complete design was simulated and synthesized using industry-standard open-source EDA tools.

2. Problem Statement

An EV charging station must ensure that charging occurs only after the vehicle is properly connected and authenticated. Incorrect charging sequences may lead to energy wastage, equipment damage, or unsafe operation.

The controller should:

- Detect vehicle connection.
- Verify authentication.
- Enable charging only after successful authentication.
- Stop charging when the battery becomes fully charged.
- Return to the idle state after charging completion or vehicle disconnection.

3. Existing System

Many basic charging systems rely on manual operation or simple relay control without proper sequencing.

Limitations

- No authentication mechanism.
- Charging may start unintentionally.
- Poor monitoring of charging status.
- Limited safety control.
- Difficult to expand with additional features.

4. Proposed System

The proposed controller is implemented using a Moore FSM.

Features

- Automatic vehicle detection.
- Authentication verification.
- Controlled charging process.
- Battery full detection.
- Safe charging termination.
- Modular RTL implementation.
- Error handling for failed authentication.

5. Objectives

- Design an EV charging controller using Verilog HDL.
- Implement FSM-based charging control.
- Verify the controller through simulation.
- Generate RTL hardware using Yosys.
- Understand complete RTL design flow.

6. System Overview

Inputs:

Signal	Description
Clk	System Clock
Reset	System Reset
vehicle_detect	Detects connected vehicle
auth_success	Authentication status
battery_full	Battery fully charged
disconnect	Vehicle disconnected

Outputs:

Signal	Description
charger_on	Enables charger
charging_done	Charging completed
auth_failed	Authentication failure

7. Working Principle

The controller continuously monitors all input signals.

1. System starts in the **IDLE** state.
2. When a vehicle is detected, the controller moves to **VEHICLE_DETECTED**.
3. If authentication succeeds, charging begins.
4. During charging, the battery status is monitored.
5. Once the battery becomes full, charging stops.
6. After the vehicle disconnects, the controller returns to **IDLE**.

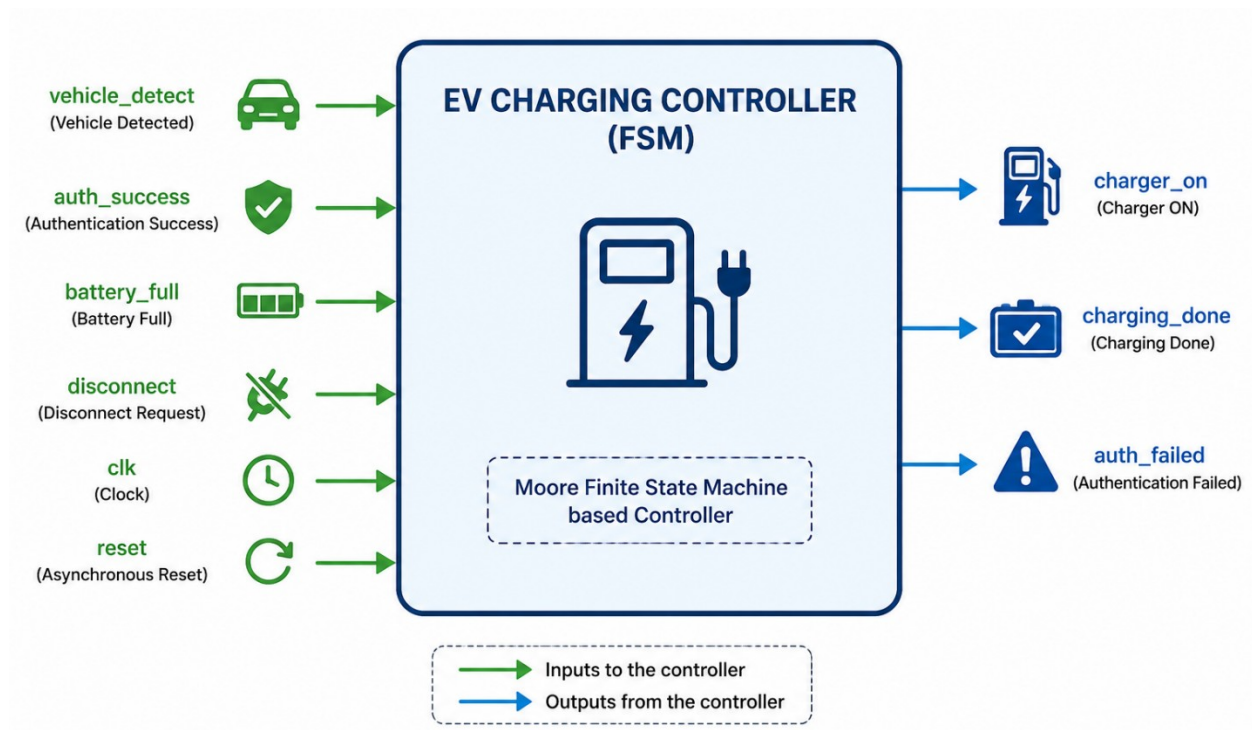
This sequential control ensures safe and reliable charging.

8. Finite State Machine (FSM)

FSM Type : Moore FSM

State	Binary	Description
IDLE	000	Waiting for vehicle
VEHICLE_DETECTED	001	Vehicle detected
AUTHENTICATION	010	Authentication in progress
CHARGING	011	Charging enabled
CHARGE_COMPLETE	100	Battery fully charged
ERROR_STATE	101	Authentication failed
DISCONNECT_STATE	110	Vehicle disconnected

9. FSM Diagram



10. State Transition Table

Current State	Condition	Next State
IDLE	vehicle_detect = 1	VEHICLE_DETECTED
VEHICLE_DETECTED	Automatic	AUTHENTICATION
AUTHENTICATION	auth_success = 1	CHARGING
AUTHENTICATION	auth_success = 0	ERROR_STATE
CHARGING	battery_full = 1	CHARGE_COMPLETE
CHARGE_COMPLETE	disconnect = 1	DISCONNECT_STATE
ERROR_STATE	disconnect = 1	IDLE
DISCONNECT_STATE	Automatic	IDLE

11. Circuit Operation

Explain each state:

IDLE

The controller waits until a vehicle is connected.

VEHICLE_DETECTED

The charger detects that a vehicle has been plugged in and waits for successful authentication.

CHARGING

Charging is enabled by activating the charger output.

CHARGE_COMPLETE

Charging is stopped after receiving the battery full signal.

DISCONNECT

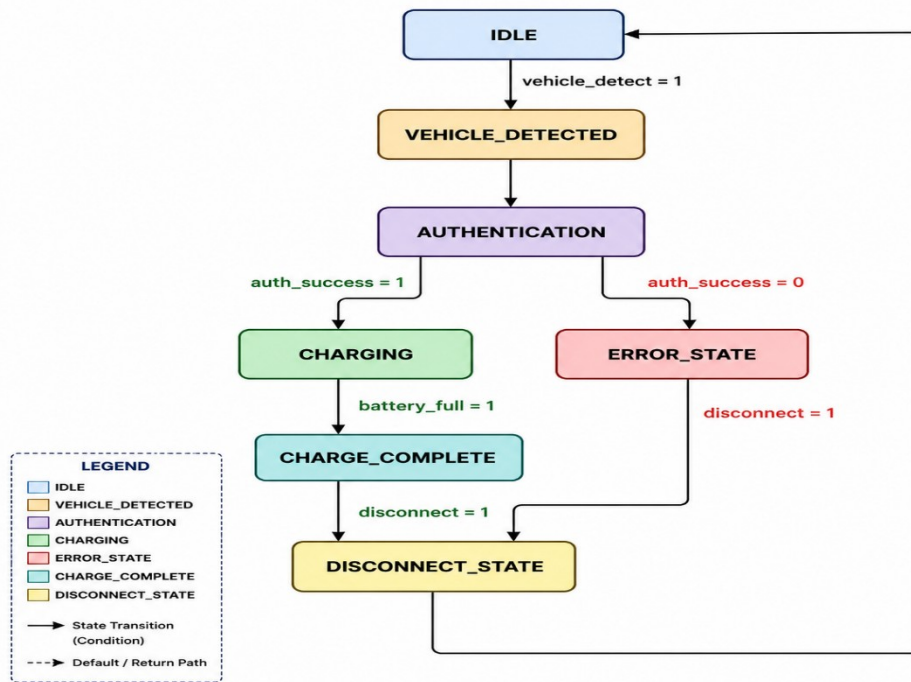
After the charging cable is removed, the controller returns to the idle state.

ERROR_STATE

If authentication fails, the controller enters the ERROR_STATE.

Charging remains disabled until the vehicle is disconnected.

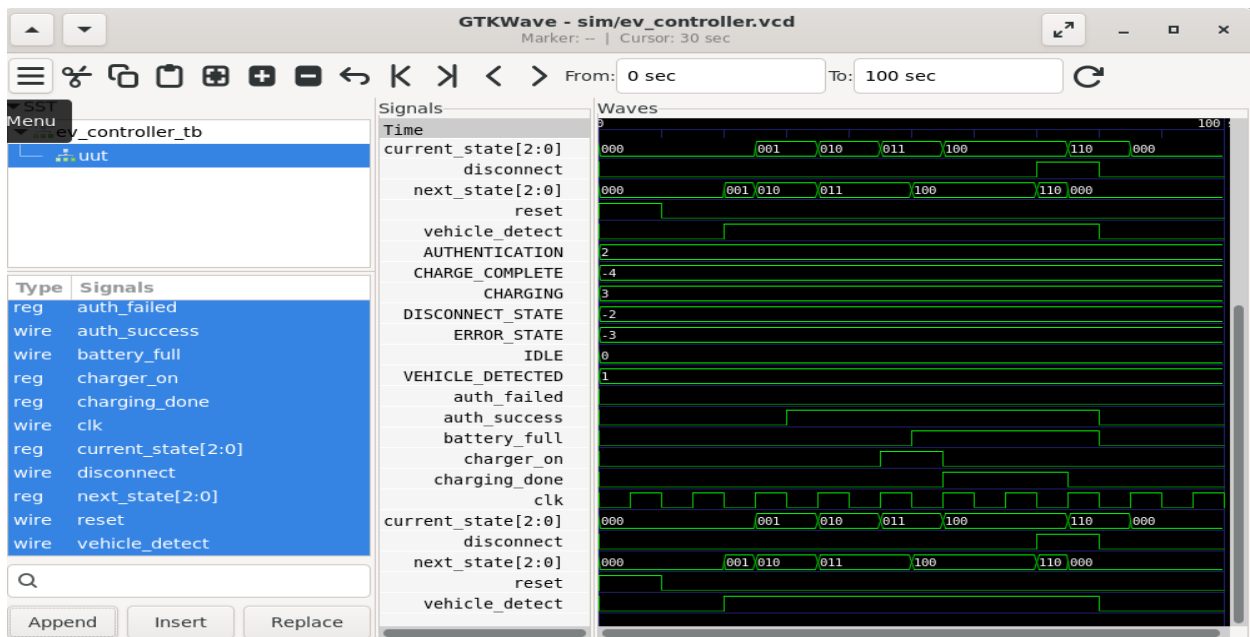
EV CHARGING CONTROLLER – FSM DIAGRAM

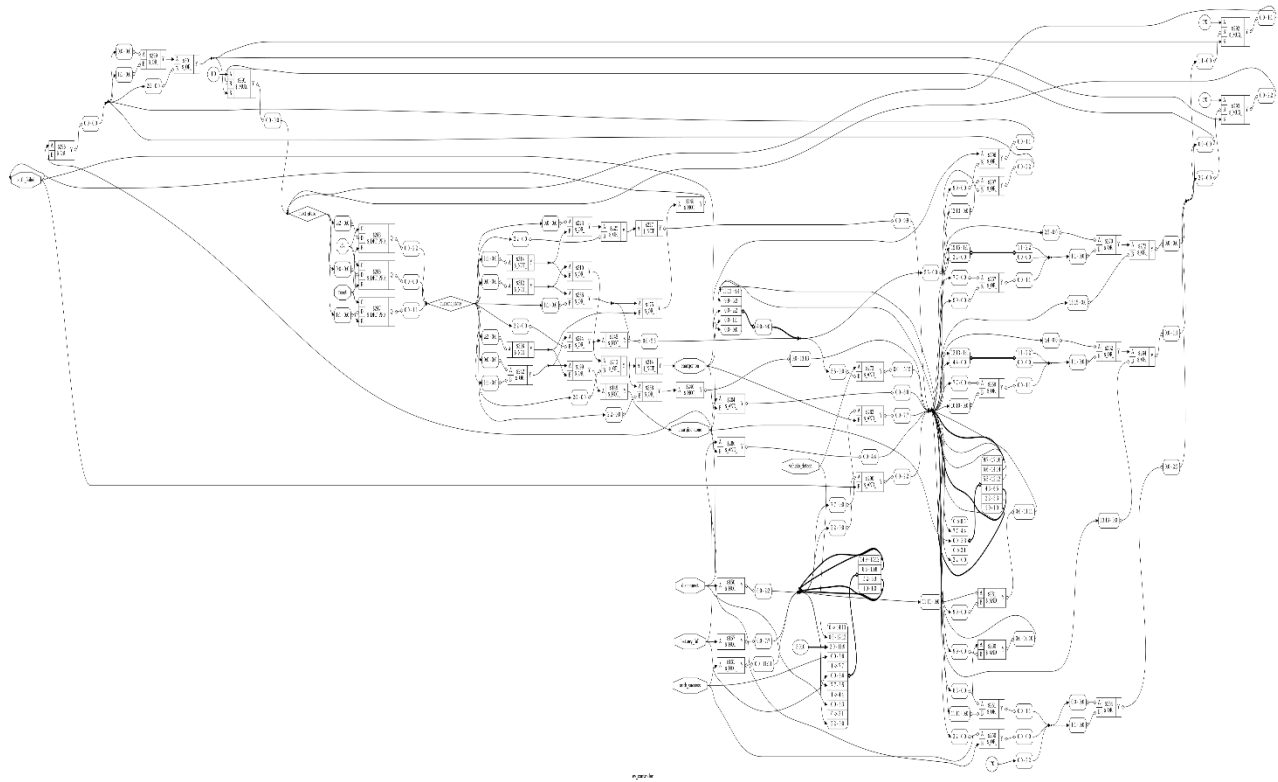


12. Testbench Design

A testbench is used to verify the functionality of the Verilog design by applying different input combinations and observing the corresponding outputs. The testbench is a non-synthesizable module that generates input stimulus for the EV Charging Station Controller.

Waveform Analysis





The waveform confirms the correct operation of the controller.

- The clock signal drives all state transitions.
- The reset signal initializes the controller to the **IDLE** state.
- Vehicle detection changes the controller to the **VEHICLE_DETECTED** state.
- Successful authentication enables charging.
- The battery full signal disables charging and indicates charging completion.
- Vehicle disconnection returns the controller to the **IDLE** state.

The waveform verifies that the controller performs all operations according to the designed FSM.

13. RTL Schematic (Yosys)

```
5. Printing statistics.
```

```
=== ev_controller ===
```

Number of wires:	22
Number of wire bits:	36
Number of public wires:	11
Number of public wire bits:	15
Number of memories:	0
Number of memory bits:	0
Number of processes:	0
Number of cells:	16
\$adff	1
\$eq	5
\$logic_not	1
\$mux	8
\$pmux	1

The RTL schematic was generated using **Yosys** after synthesizing the Verilog design. It illustrates the hardware implementation of the controller, including the FSM logic, state registers, and output control circuitry. The schematic confirms that the Verilog code is successfully synthesized into digital hardware.

Logic Utilization

The design was synthesized using Yosys at the RTL level. The synthesized circuit consists of state registers and combinational logic required for implementing the finite state machine. Since the design was not mapped to a specific FPGA device, FPGA-specific LUT utilization was not generated.

Advantages

- Safe and controlled charging operation.
- Prevents unauthorized charging.

- Simple Moore FSM implementation.
- Easy to simulate and verify.
- Modular and easy to enhance.

Applications

- Public EV charging stations
- Home EV charging systems
- Smart charging infrastructure
- FPGA and VLSI learning projects
- Electric vehicle control systems

Tools Used

- **Verilog HDL** – Used for designing the RTL of the EV Charging Station Controller.
- **Icarus Verilog (iverilog)** – Used for compiling the Verilog source code and testbench.
- **VVP (Verilog Virtual Processor)** – Used to execute the compiled simulation.
- **GTKWave** – Used to view and analyze the simulation waveform (.vcd file).
- **Yosys** – Used for RTL synthesis and generation of the RTL schematic.
- **Ubuntu (WSL)** – Used as the Linux development environment for RTL design and simulation.
- **Nano Editor** – Used for writing and editing Verilog source files and scripts.
- **Git** – Used for version control and tracking project changes.
- **GitHub** – Used to host and manage the project repository.
- **Graphviz** – Used by Yosys to generate the RTL schematic visualization.

Future Scope

- Add **RFID-based authentication** for secure user verification.
- Implement **fast charging mode** with intelligent charging control.
- Integrate **UART communication** for monitoring and debugging.
- Add **CAN bus communication** for interaction with electric vehicle systems.
- Implement **battery health monitoring** and diagnostics.
- Support **multiple charging ports** for simultaneous EV charging.
- Integrate **IoT-based remote monitoring** and control.

- Add **smart charging algorithms** for improved energy efficiency.
- Implement **fault detection and protection mechanisms** such as overcurrent and overvoltage protection.
- Deploy the design on an **FPGA development board** for real-time hardware validation.

14. Conclusion

The EV Charging Station Controller was successfully designed using Verilog HDL and implemented using a Moore Finite State Machine. The design was verified through simulation using Icarus Verilog and GTKWave, and the RTL schematic was generated using Yosys. The results confirm that the controller performs safe and reliable charging operations, making it suitable as a basic digital controller for EV charging applications.